

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Paper No. CHB-201: Inorganic & Physical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-I – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Give a valid reason for the high solubility of BeF_2 in water, while all other alkaline earth fluorides are insoluble.
- (b) Write the structures of BeCl_2 in both the vapour phase and solid state.
- (c) Write the molecular formula of 'Borax' and draw its structure.
- (d) Write the formula of Beryl. To which class of silicates does it belong?
- (e) Name the anhydride of HNO_3 and draw its gas-phase structure.

Question 2.

- (a) State the Kapustinskii equation for lattice energy and calculate the lattice energy (ΔH_L) of NaCl using a Born-Haber cycle with the following data (all values in kJ/mol):

$$\Delta H_f^\circ = -411.2, \quad \Delta H_{\text{sub}} = 108.4, \quad \text{IE} = 495.4, \quad \Delta H_{\text{diss}}(\text{Cl}_2) = 241.8, \quad \text{EA}(\text{Cl}) = -348.6$$

[6]

- (b) Define 'electrochemical series'. Answer the following, giving reasons, based on the standard reduction potentials (E° in volts):

$$\text{Zn}^{2+}/\text{Zn} = -0.76, \quad \text{Fe}^{2+}/\text{Fe} = -0.44, \quad \text{Cu}^{2+}/\text{Cu} = +0.34, \quad \text{Ag}^+/\text{Ag} = +0.80$$

- (i) Can we store copper sulphate solution in a zinc container?
- (ii) Which is the strongest reducing agent among these metals?

[6]

Question 3.

- (a) Answer the following with respect to diborane (B_2H_6):

- (i) Name the experimental or physical property which confirmed that rotation between the two boron atoms is hindered.
- (ii) Give two experimental evidences which support that four terminal hydrogen atoms are in a different environment from the two bridging hydrogen atoms.

[6]

- (b) Draw the structures of hydrazine (N_2H_4) and hydroxylamine (NH_2OH). Explain why hydrazine is a weaker base than ammonia.

[6]

Question 4.

- (a) Show hybrid orbitals, draw structures, and mention the shapes of the following: (i) XeO_2F_2 and (ii) XeO_3F_2 .
- (b) How do you name the $[\text{XeO}_6]^{4-}$ ion? Show hybrid orbitals, mention the number of σ - and π -bonds, and draw the structure of the ion.

[6]

[6]

SECTION B

(Physical Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) What is the influence of temperature on the surface tension and viscosity of a liquid?
- (b) Define critical temperature (T_c), Boyle temperature (T_B), and inversion temperature (T_i).
- (c) Heat change (q) is a path-dependent thermodynamic quantity. State the conditions under which it becomes a state function.
- (d) Show that for a reversible adiabatic expansion of an ideal gas, the work done is given by $W = C_v(T_1 - T_2)$.
- (e) Define the term 'mean free path' of a gas molecule.

Question 6.

- (a) Derive Kirchhoff's equations for the variation of enthalpy of reaction with temperature. [6]
- (b) Show that the entropy change in an isothermal reversible process for the universe is zero, while it is positive in a spontaneous (irreversible) process. [6]

Question 7.

- (a) For reversible isothermal and adiabatic expansion of an ideal gas, demonstrate how thermodynamic quantities (w , q , ΔU , and ΔH) can be estimated. [6]
- (b) Using the kinetic gas equation, deduce Avogadro's law and Graham's law of diffusion. [6]

Question 8.

- (a) Define the coefficient of viscosity of a liquid. Discuss the Ostwald viscometer method for the determination of the viscosity of a liquid. [6]
- (b) Write the van der Waals equation of state for a real gas. Define critical constants (P_c , V_c , T_c) in terms of van der Waals constants (a and b). [6]

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